5.1

Let $f : A \to B$ be an integral homomorphism of rings. Show that $f^* : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is a **closed mapping**, i.e. that it maps closed sets to closed sets. (This is a geometrical equivalent of (5.10).)
Proof: Factor f into $A \xrightarrow{g} C = f(A) \overset{\iota}{\hookrightarrow} B$. Then since g is surjective, g^* is a closed map. We show that $\iota^* : \mathrm{Spec}(C) \to \mathrm{Spec}(B)$ is a closed mapping.
For any closed set $V(\mathfrak{b}) \subset \mathrm{Spec}(B)$, we have $f^*(V(\mathfrak{b})) = V(\mathfrak{b}^c)$: If $\mathfrak{q} \in V(\mathfrak{b})$ then $\mathfrak{b} \subset \mathfrak{q} \implies \mathfrak{b}^c \subset \mathfrak{q}^c$ so $f^*(V(\mathfrak{b})) \subset V(\mathfrak{b}^c)$. If $\mathfrak{p} \in \mathrm{Spec}(C)$ then since $C \subset B$ and B is integral over C, by Theorem5.10 there exists $\mathfrak{q} \in \mathrm{Spec}(B)$ such that $\mathfrak{p} = \mathfrak{q} \cap C$. $\mathfrak{b}^c = \mathfrak{b} \cap C \subset \mathfrak{p} = \mathfrak{q} \cap C$ implies $\mathfrak{q} \in V(\mathfrak{b})$, and $f^*(\mathfrak{q}) = \mathfrak{p}$. Hence $f^*(V(\mathfrak{b})) = V(\mathfrak{b}^c)$ and f^* is a closed mapping.

5.2

Let A be a subring of a ring B such that B is integral over A, and let $f : A \to \Omega$ be a homomorphism of A into an algebraically closed field Ω. Show that f can be extended to a homomorphism of B into Ω.
Proof: Let $\Sigma = \{(C, g) : A \subset C \subset B, g : C \to \Omega \text{ homomorphism of rings}, g|_A = f\}$, and consider the partial ordering $(C, g) \leqslant (C', g')$ if $C \subset C'$ and $g'|_C = g$. Clearly every chain in Σ has an upper bound. By Zorn's lemma we can take a maximal element $(C_0, g_0) \in \Sigma$.
If $C_0 = B$ we are done. Otherwise take $b \in B \backslash C_0$, and let $C_1 = C_0[b]$. B is integral over A hence also over C, and suppose b is a root of $p(x) = x^n + a_1 x^{n-1} + \cdots + a_{n-1}x + a_n = 0$. Since Ω is algebraically closed, take a root $y \in \Omega$ of $y^n + f(a_1)y^{n-1} + \cdots + f(a_n) = 0$. Consider the homomorphism $\tilde{g} : C_0[x] \to \Omega$ induced by $x \mapsto y$, then $(p) \subset \mathrm{Ker}\tilde{g}$ so $\tilde{g}$ induces a homomorphism $g_1 : C_1 = C_0[x]/(p) \to \Omega$ such that $g_1(b) = y$ and $g_1|_{C_0} = g_0$, a contradiction.

5.3

Let $f : B \to B'$ be a homomorphism of A-algebras, and let C be an A-algebra. If f is integral, prove that $f \otimes 1 : B \otimes_A C \to B' \otimes_A C$ is integral. (This includes (5.6) ii) as a special case.)
Proof: For any $b' \otimes c \in B' \otimes_A C$, since b' is integral over $f(B)$, suppose $(b')^n + f(b_1)(b')^{n-1} + \cdots + f(b_n) = 0$. Note that

$$(b' \otimes c)^n + (f(b_1) \otimes c) \cdot (b' \otimes c)^{n-1} + \cdots + (f(b_n) \otimes c^n) = 0$$

where $f(b_k) \otimes c^k \in \mathrm{Im}(f \otimes 1)$. Hence $b' \otimes c$ is integral over $\mathrm{Im}(f \otimes 1)$. Since every element of $B' \otimes_A C$ is a finite sum of $b' \otimes c$, $f \otimes 1$ is integral.

5.4

Let A be a subring of a ring B such that B is integral over A. Let $\mathfrak{n}$ be a maximal ideal of B and let $\mathfrak{m} = \mathfrak{n} \cap A$ be the corresponding maximal ideal of A. Is $B_\mathfrak{n}$ necessarily integral over $A_\mathfrak{m}$?
Solution: No. Let $A = k[x^2] \subset B = k[x]$, where k is a field, then x is the root of $t^2 - x^2 = 0$ so B is integral over A. Let $\mathfrak{n} = (x - 1)$ and $\mathfrak{m} = \mathfrak{n} \cap A = (x^2 - 1)$, then $(x + 1) \notin \mathfrak{n}$ so $\frac{1}{x+1} \in B_\mathfrak{n}$. The minimal polynomial of $\frac{1}{x+1}$ over $k(x^2)$ is $t^2 + \frac{2}{x^2-1}t - \frac{1}{x^2-1} = 0$. However, $A_\mathfrak{m}$ is a DVR, hence integrally closed, so $\frac{1}{x+1}$ is not integral over $A_\mathfrak{m}$.

5.5

Let $A \subseteq B$ be rings, B integral over A.
i) If $x \in A$ is a unit in B then it is a unit in A.
ii) The Jacobson radical of A is the contraction of the Jacobson radical of B.
Proof: (i) Suppose x^{-1} has a minimal polynomial $t^n + a_1 t^{n-1} + \cdots + a_n = 0$, then $1 = x(-a_1 - a_2 x - \cdots - a_n x^{n-1})$ where $-a_1 - a_2 x - \cdots - a_n x^{n-1} \in A$, hence x is a unit in A.
(ii) By Corollary5.8 and going-up theorem, every maximal ideal $\mathfrak{m}$ in A gives a unique maximal ideal $\mathfrak{n} \cap A = \mathfrak{m}$. Hence $\mathfrak{R}(A) = \mathfrak{R}(B)^c$.

5.6

Let $B_1, \ldots, B_n$ be integral A-algebras. Show that $\prod_{i=1}^n B_i$ is an integral A-algebra.
Proof: We prove the case $n = 2$: If B, C are integral $A-$algebras, then for any element $(b, c) \in B \times C$, suppose monic polynomials $f, g \in A[x]$ annihilate b, c, then $fg \in A[x]$ is a monic polynomial that annihilates (b, c). Hence $B \times C$ is integral over A.

5.7

Let A be a subring of a ring B, such that the set $B - A$ is closed under multiplication. Show that A is integrally closed in B.
Proof: Suppose $b \in \overline{A} \backslash A$, and $b^n + a_1 b^{n-1} + \cdots + a_n = 0$ where $a_i \in A$. Take the minimal such n, then $b(b^{n-1} + a_1 b^{n-2} + \cdots + a_{n-1}) = -a_n \in A$, where $b \notin A$. Hence $a = b^{n-1} + a_1 b^{n-2} + \cdots + a_{n-1} \in A$, so $b^{n-1} + a_1 b^{n-2} + \cdots + (a_{n-1} - a) = 0$, contradicting minimality of n. Hence $\overline{A} = A$.

5.8

i) Let A be a subring of an integral domain B, and let C be the integral closure of A in B. Let f, g be monic polynomials in $B[x]$ such that $fg \in C[x]$. Then f, g are in $C[x]$.
ii) Prove the same result without assuming that B (or A) is an integral domain.
Proof: (i) WLOG assume $A = C$ be integrally closed. Consider the splitting field L of fg over $K = \mathrm{Frac}(B)$, and suppose $f = \prod_{i=1}^n (x - \alpha_i)$, $g = \prod_{j=1}^m (x - \beta_j)$, then each α_i, β_j is a root of $fg \in C[x]$, hence integral in L over C. Therefore the coefficients of f, g are integral over C. Since the coefficients are in B and C is integrally closed in B, $f, g \in C[x]$.
(ii) Following (i), we want to create a "splitting" ring $R \supset B$ such that fg splits in $R[x]$, then analogous to (i) we have $f, g \in C[x]$.
We prove by induction on $n = \deg f$ that for monic $f \in B[x]$ there exists $R \supset B$ such that f splits in $R[x]$. The base $n = 1$ is trivial. Suppose it holds for $n - 1$, then consider $B' = B[t]/(f(t))$, we obtain a root t of f in B'. Hence $f = (x - t) f'$ for some monic $f' \in B'[x]$, and $\deg f' = n - 1$. By induction hypothesis there exists $R \supset B'$ such that f' splits in $R[x]$. Then f splits in $R[x]$.

5.9

Let A be a subring of a ring B and let C be the integral closure of A in B. Prove that $C[x]$ is the integral closure of $A[x]$ in $B[x]$.
Proof: Clearly $A[x] \subset C[x] \subset \overline{A[x]}$, so it suffices to show $\overline{C[x]} = C[x]$.
On first thoughts, given $f \in \overline{C[x]}$ and $f^n + g_1 f^{n-1} + \cdots + g_n = 0$ where $g_i \in C[x]$, we have $f(f^{n-1} + g_1 f^{n-2} + \cdots + g_{n-1}) = -g_n \in C[x]$, so by Ex5.8 $f \in C[x]$. However, f and $f^{n-1} + \cdots + g_{n-1}$ may not be monic.
A revised proof: Take r larger than n and $\max \deg g_i$, and $f_1 = f - x^r$, then expanding $(f_1 + x^r)^n + g_1(f_1 + x^r)^{n-1} + \cdots + g_n = 0$ we obtain $f_1^n + h_1 f_1^{n-1} + \cdots + h_n = 0$, where $\deg f_1 = r$, $h_i \in C[x]$ and f_1, h_n are monic. Since $f_1(f_1^{n-1} + h_1 f_1^{n-2} + \cdots + h_{n-1}) = -h_n \in C[x]$, and f_1, h_n are monic, by Ex5.8 $f_1 \in C[x]$. Hence $f = f_1 + x^r \in C[x]$, and $C[x] = \overline{A[x]}$.

5.10

A ring homomorphism $f : A \to B$ is said to have the **going-up property** (resp. the **going-down property**) if the conclusion of the going-up theorem (5.11) (resp. the going-down theorem (5.16)) holds for B and its subring $f(A)$.
Let $f^* : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ be the mapping associated with f.
i) Consider the following three statements:
(a) f^* is a closed mapping.
(b) f has the going-up property.
(c) Let $\mathfrak{q}$ be any prime ideal of B and let $\mathfrak{p} = \mathfrak{q}^c$. Then $f^* : \mathrm{Spec}(B/\mathfrak{q}) \to \mathrm{Spec}(A/\mathfrak{p})$ is surjective.
Prove that (a) $\Rightarrow$ (b) $\Leftrightarrow$ (c). (See also Chapter 6, Exercise 11.)
ii) Consider the following three statements:
(a′) f^* is an open mapping.
(b′) f has the going-down property.
(c′) For any prime ideal $\mathfrak{q}$ of B, if $\mathfrak{p} = \mathfrak{q}^c$, then $f^* : \mathrm{Spec}(B_\mathfrak{q}) \to \mathrm{Spec}(A_\mathfrak{p})$ is surjective.
Prove that (a′) $\Rightarrow$ (b′) $\Leftrightarrow$ (c′). (See also Chapter 7, Exercise 23.)[To prove that (a′) $\Rightarrow$ (c′), observe that $B_\mathfrak{q}$ is the direct limit of the rings B_t where $t \in B - \mathfrak{q}$; hence, by Chapter 3, Exercise 26, we have $f^*(\mathrm{Spec}(B_\mathfrak{q})) = \bigcap_t f^*(\mathrm{Spec}(B_t)) = \bigcap_t f^*(Y_t)$. Since Y_t is an open neighborhood of $\mathfrak{q}$ in Y, and since f^* is open, it follows that $f^*(Y_t)$ is an open neighborhood of $\mathfrak{p}$ in X and therefore contains $\mathrm{Spec}(A_\mathfrak{p})$.]

5.11

Let $f : A \to B$ be a flat homomorphism of rings. Then f has the going-down property. [Chapter 3, Exercise 18.]

5.12

Let G be a finite group of automorphisms of a ring A, and let A^G denote the subring of G-invariants, that is of all $x \in A$ such that $\sigma(x) = x$ for all $\sigma \in G$. Prove that A is integral over A^G.[If $x \in A$, observe that x is a root of the polynomial $\prod_{\sigma \in G}(t - \sigma(x))$.]
Let S be a multiplicatively closed subset of A such that $\sigma(S) \subseteq S$ for all $\sigma \in G$, and let $S^G = S \cap A^G$. Show that the action of G on A extends to an action on $S^{-1}A$, and that $(S^G)^{-1}A^G \cong (S^{-1}A)^G$.

5.13

In the situation of Exercise 12, let $\mathfrak{p}$ be a prime ideal of A^G, and let P be the set of prime ideals of A whose contraction is $\mathfrak{p}$. Show that G acts transitively on P. In particular, P is finite.[Let $\mathfrak{p}_1, \mathfrak{p}_2 \in P$ and let $x \in \mathfrak{p}_1$. Then $\prod_\sigma \sigma(x) \in \mathfrak{p}_1 \cap A^G = \mathfrak{p} \subseteq \mathfrak{p}_2$, hence $\sigma(x) \in \mathfrak{p}_2$ for some $\sigma \in G$. Deduce that $\mathfrak{p}_1$ is contained in $\bigcup_{\sigma \in G} \sigma(\mathfrak{p}_2)$, and then apply (1.11) and (5.9).]

5.14

Let A be an integrally closed domain, K its field of fractions and L a finite normal separable extension of K. Let G be the Galois group of L over K and let B be the integral closure of A in L. Show that $\sigma(B) = B$ for all $\sigma \in G$, and that $A = B^G$.

5.15

Let A, K be as in Exercise 14, let L be any finite extension field of K, and let B be the integral closure of A in L. Show that, if $\mathfrak{p}$ is any prime ideal of A, then the set of prime ideals $\mathfrak{q}$ of B which contract to $\mathfrak{p}$ is finite (in other words, that $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ has finite fibers).[Reduce to the two cases (a) L separable over K and (b) L purely inseparable over K. In case (a), embed L in a finite normal separable extension of K, and use Exercises 13 and 14. In case (b), if $\mathfrak{q}$ is a prime ideal of B such that $\mathfrak{q} \cap A = \mathfrak{p}$, show that $\mathfrak{q}$ is the set of all $x \in B$ such that $x^{p^m} \in \mathfrak{p}$ for some $m \geqslant 0$, where p is the characteristic of K, and hence that $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is bijective in this case.]

Noether's normalization lemma

5.16

Let k be a field and let $A \neq 0$ be a finitely generated k-algebra. Then there exist elements $y_1, \ldots, y_r \in A$ which are algebraically independent over k and such that A is integral over $k[y_1, \ldots, y_r]$.
We shall assume that k is infinite. (The result is still true if k is finite, but a different proof is needed.) Let $x_1, \ldots, x_n$ generate A as a k-algebra. We can renumber the x_i so that $x_1, \ldots, x_r$ are algebraically independent over k and each of $x_{r+1}, \ldots, x_n$ is algebraic over $k[x_1, \ldots, x_r]$. Now proceed by induction on n. If $n = r$ there is nothing to do, so suppose $n > r$ and the result true for $n - 1$ generators. The generator x_n is algebraic over $k[x_1, \ldots, x_{n-1}]$, hence there exists a polynomial $f \neq 0$ in n variables such that $f(x_1, \ldots, x_{n-1}, x_n) = 0$. Let F be the homogeneous part of highest degree in f. Since k is infinite, there exist $\lambda_1, \ldots, \lambda_{n-1} \in k$ such that $F(\lambda_1, \ldots, \lambda_{n-1}, 1) \neq 0$. Put $x_i' = x_i - \lambda_i x_n$ $(1 \leqslant i \leqslant n-1)$. Show that x_n is integral over the ring $A' = k[x_1', \ldots, x_{n-1}']$, and hence that A is integral over A'. Then apply the inductive hypothesis to A' to complete the proof.
From the proof it follows that $y_1, \ldots, y_r$ may be chosen to be linear combinations of $x_1, \ldots, x_n$. This has the following geometrical interpretation: if k is algebraically closed and X is an affine algebraic variety in k^n with coordinate ring $A \neq 0$, then there exists a linear subspace L of dimension r in k^n and a linear mapping of k^n onto L which maps X onto L. [Use Exercise 2.]

Nullstellensatz (weak form).

5.17

Let X be an affine algebraic variety in k^n, where k is an algebraically closed field, and let $I(X)$ be the ideal of X in the polynomial ring $k[t_1, \ldots, t_n]$ (Chapter 1, Exercise 27). If $I(X) \neq (1)$ then X is not empty. [Let $A = k[t_1, \ldots, t_n]/I(X)$ be the coordinate ring of X. Then $A \neq 0$, hence by Exercise 16 there exists a linear subspace L of dimension $\geqslant 0$ in k^n and a mapping of X onto L. Hence $X \neq \varnothing$.]
Deduce that every maximal ideal in the ring $k[t_1, \ldots, t_n]$ is of the form $(t_1 - a_1, \ldots, t_n - a_n)$ where $a_i \in k$.

5.18

Let k be a field and let B be a finitely generated k-algebra. Suppose that B is a field. Then B is a finite algebraic extension of k. (This is another version of Hilbert's Nullstellensatz. The following proof is due to Zariski. For other proofs, see (5.24), (7.9).)
Let $x_1, \ldots, x_n$ generate B as a k-algebra. The proof is by induction on n. If $n = 1$ the result is clearly true, so assume $n > 1$. Let $A = k[x_1]$ and let $K = k(x_1)$ be the field of fractions of A. By the inductive hypothesis, B is a finite algebraic extension of K, hence each of $x_2, \ldots, x_n$ satisfies a monic polynomial equation with coefficients in K, i.e. coefficients of the form a/b where a and b are in A. If f is the product of the denominators of all these coefficients, then each of $x_2, \ldots, x_n$ is integral over A_f. Hence B and therefore K is integral over A_f.
Suppose x_1 is transcendental over k. Then A is integrally closed, because it is a unique factorization domain. Hence A_f is integrally closed (5.12), and therefore $A_f = K$, which is clearly absurd. Hence x_1 is algebraic over k, hence K (and therefore B) is a finite extension of k.

5.19

Deduce the result of Exercise 17 from Exercise 18.

5.20

Let A be a subring of an integral domain B such that B is finitely generated over A. Show that there exists $s \neq 0$ in A and elements $y_1, \ldots, y_n$ in B, algebraically independent over A and such that B_s is integral over B'_s, where $B' = A[y_1, \ldots, y_n]$.[Let $S = A - \{0\}$ and let $K = S^{-1}A$, the field of fractions of A. Then $S^{-1}B$ is a finitely generated K-algebra and therefore by the normalization lemma (Exercise 16) there exist $x_1, \ldots, x_n$ in $S^{-1}B$, algebraically independent over K and such that $S^{-1}B$ is integral over $K[x_1, \ldots, x_n]$. Let $z_1, \ldots, z_m$ generate B as an A-algebra. Then each z_j (regarded as an element of $S^{-1}B$) is integral over $K[x_1, \ldots, x_n]$. By writing an equation of integral dependence for each z_j, show that there exists $s \in S$ such that $x_i = y_i/s$ ($1 \leqslant i \leqslant n$) with $y_i \in B$, and such that each sz_j is integral over B'. Deduce that this s satisfies the conditions stated.]

5.21

Let A, B be as in Exercise 20. Show that there exists $s \neq 0$ in A such that, if Ω is an algebraically closed field and $f : A \to \Omega$ is a homomorphism for which $f(s) \neq 0$, then f can be extended to a homomorphism $B \to \Omega$.[With the notation of Exercise 20, f can be extended first of all to B', for example by mapping each y_i to 0; then to B'_s (because $f(s) \neq 0$), and finally to B_s (by Exercise 2, because B_s is integral over B'_s).]

5.22

Let A, B be as in Exercise 20. If the Jacobson radical of A is zero, then so is the Jacobson radical of B.[Let $v \neq 0$ be an element of B. We have to show that there is a maximal ideal of B which does not contain v. By applying Exercise 21 to the ring B_v and its subring A, we obtain an element $s \neq 0$ in A. Let $\mathfrak{m}$ be a maximal ideal of A such that $s \notin \mathfrak{m}$, and let $k = A/\mathfrak{m}$. Then the canonical mapping $A \to k$ extends to a homomorphism g of B_v into an algebraic closure Ω of k. Show that $g(v) \neq 0$ and that $\mathrm{Ker}(g) \cap B$ is a maximal ideal of B.]

5.23

Let A be a ring. Show that the following are equivalent:
i) Every prime ideal in A is an intersection of maximal ideals.
ii) In every homomorphic image of A the nilradical is equal to the Jacobson radical.
iii) Every prime ideal in A which is not maximal is equal to the intersection of the prime ideals which contain it strictly.[The only hard part is iii) $\Rightarrow$ ii). Suppose ii) false, then there is a prime ideal which is not an intersection of maximal ideals. Passing to the quotient ring, we may assume that A is an integral domain whose Jacobson radical $\mathfrak{R}$ is not zero. Let f be a non-zero element of $\mathfrak{R}$. Then $A_f \neq 0$, hence A_f has a maximal ideal, whose contraction in A is a prime ideal $\mathfrak{p}$ such that $f \notin \mathfrak{p}$, and which is maximal with respect to this property. Then $\mathfrak{p}$ is not maximal and is not equal to the intersection of the prime ideals strictly containing $\mathfrak{p}$.]
A ring A with the three equivalent properties above is called a *Jacobson ring*.

5.24

Let A be a Jacobson ring (Exercise 23) and B an A-algebra. Show that if B is either (i) integral over A or (ii) finitely generated as an A-algebra, then B is Jacobson. [Use Exercise 22 for (ii).]
In particular, every finitely generated ring, and every finitely generated algebra over a field, is a Jacobson ring.

5.25

Let A be a ring. Show that the following are equivalent:
i) A is a Jacobson ring;
ii) Every finitely generated A-algebra B which is a field is finite over A.[i) $\Rightarrow$ ii). Reduce to the case where A is a subring of B, and use Exercise 21. If $s \in A$ is as in Exercise 21, then there exists a maximal ideal $\mathfrak{m}$ of A not containing s, and the homomorphism $A \to A/\mathfrak{m} = k$ extends to a homomorphism g of B into the algebraic closure of k. Since B is a field, g is injective, and $g(B)$ is algebraic over k, hence finite algebraic over k.
ii) $\Rightarrow$ i). Use criterion iii) of Exercise 23. Let $\mathfrak{p}$ be a prime ideal of A which is not maximal, and let $B = A/\mathfrak{p}$. Let f be a non-zero element of B. Then B_f is a finitely generated A-algebra. If it is a field it is finite over B, hence integral over B and therefore B is a field by (5.7). Hence B_f is not a field and therefore has a non-zero prime ideal, whose contraction in B is a non-zero ideal $\mathfrak{p}'$ such that $f \notin \mathfrak{p}'$.]

5.26

Let X be a topological space. A subset of X is *locally closed* if it is the intersection of an open set and a closed set, or equivalently if it is open in its closure.
The following conditions on a subset X_0 of X are equivalent:
(1) Every non-empty locally closed subset of X meets X_0;
(2) For every closed set E in X we have $\overline{E \cap X_0} = E$;
(3) The mapping $U \mapsto U \cap X_0$ of the collection of open sets of X onto the collection of open sets of X_0 is bijective.
A subset X_0 satisfying these conditions is said to be *very dense* in X.
If A is a ring, show that the following are equivalent:
i) A is a Jacobson ring;
ii) The set of maximal ideals of A is very dense in $\mathrm{Spec}(A)$;
iii) Every locally closed subset of $\mathrm{Spec}(A)$ consisting of a single point is closed.[ii) and iii) are geometrical formulations of conditions ii) and iii) of Exercise 23.]

Valuation rings and valuations

5.27

Let A, B be two local rings. B is said to **dominate** A if A is a subring of B and the maximal ideal $\mathfrak{m}$ of A is contained in the maximal ideal $\mathfrak{n}$ of B (or, equivalently, if $\mathfrak{m} = \mathfrak{n} \cap A$). Let K be a field and let Σ be the set of all local subrings of K. If Σ is ordered by the relation of domination, show that Σ has maximal elements and that $A \in \Sigma$ is maximal if and only if A is a valuation ring of K. [Use (5.21).]

5.28

Let A be an integral domain, K its field of fractions. Show that the following are equivalent:
(1) A is a valuation ring of K;
(2) If $\mathfrak{a}, \mathfrak{b}$ are any two ideals of A, then either $\mathfrak{a} \subseteq \mathfrak{b}$ or $\mathfrak{b} \subseteq \mathfrak{a}$.
Deduce that if A is a valuation ring and $\mathfrak{p}$ is a prime ideal of A, then $A_{\mathfrak{p}}$ and $A/\mathfrak{p}$ are valuation rings of their fields of fractions.

5.29

Let A be a valuation ring of a field K. Show that every subring of K which contains A is a local ring of A.

5.30

Let A be a valuation ring of a field K. The group U of units of A is a subgroup of the multiplicative group K^* of K.
Let $\Gamma = K^*/U$. If $\xi, \eta \in \Gamma$ are represented by $x, y \in K$, define $\xi \geqslant \eta$ to mean $xy^{-1} \in A$. Show that this defines a total ordering on Γ which is compatible with the group structure (i.e., $\xi \geqslant \eta \Rightarrow \xi\omega \geqslant \eta\omega$ for all $\omega \in \Gamma$). In other words, Γ is a totally ordered abelian group. It is called the *value group* of A.
Let $v : K^* \to \Gamma$ be the canonical homomorphism. Show that $v(x+y) \geqslant \min(v(x), v(y))$ for all $x, y \in K^*$.

5.31

Conversely, let Γ be a totally ordered abelian group (written *additively*), and let K be a field. A *valuation* of K with values in Γ is a mapping $v : K^* \to \Gamma$ such that
(1) $v(xy) = v(x) + v(y)$,
(2) $v(x+y) \geqslant \min(v(x), v(y))$,
for all $x, y \in K^*$. Show that the set of elements $x \in K^*$ such that $v(x) \geqslant 0$ is a valuation ring of K. This ring is called the *valuation ring* of v, and the subgroup $v(K^*)$ of Γ is the *value group* of v.
Thus the concepts of valuation ring and valuation are essentially equivalent.

5.32

Let Γ be a totally ordered abelian group. A subgroup Δ of Γ is *isolated* in Γ if, whenever $0 \leqslant \beta \leqslant \alpha$ and $\alpha \in \Delta$, we have $\beta \in \Delta$. Let A be a valuation ring of a field K, with value group Γ (Exercise 31). If $\mathfrak{p}$ is a prime ideal of A, show that $v(A - \mathfrak{p})$ is the set of elements $\geqslant 0$ in an isolated subgroup Δ of Γ, and that the mapping so defined of $\operatorname{Spec}(A)$ into the set of isolated subgroups of Γ is bijective.
If $\mathfrak{p}$ is a prime ideal of A, what are the value groups of the valuation rings $A/\mathfrak{p}$, $A_{\mathfrak{p}}$?

5.33

Let Γ be a totally ordered abelian group. We shall show how to construct a field K and a valuation v of K with Γ as value group. Let k be any field and let $A = k[\Gamma]$ be the group algebra of Γ over k. By definition, A is freely generated as a k-vector space by elements x_α ($\alpha \in \Gamma$) such that $x_\alpha x_\beta = x_{\alpha+\beta}$. Show that A is an integral domain.
If $u = \lambda_1 x_{\alpha_1} + \cdots + \lambda_n x_{\alpha_n}$ is any non-zero element of A, where the λ_i are all $\neq 0$ and $\alpha_1 < \cdots < \alpha_n$, define $v_0(u)$ to be α_1. Show that the mapping $v_0 : A - \{0\} \to \Gamma$ satisfies conditions (1) and (2) of Exercise 31.
Let K be the field of fractions of A. Show that v_0 can be uniquely extended to a valuation v of K, and that the value group of v is precisely Γ.

5.34

Let A be a valuation ring and K its field of fractions. Let $f : A \to B$ be a ring homomorphism such that $f^* : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is a **closed mapping**. Then if $g : B \to K$ is any A-algebra homomorphism (i.e., if $g \circ f$ is the embedding of A in K) we have $g(B) = A$.[Let $C = g(B)$; obviously $C \supseteq A$. Let $\mathfrak{n}$ be a maximal ideal of C. Since f^* is closed, $\mathfrak{m} = \mathfrak{n} \cap A$ is the maximal ideal of A, whence $A_{\mathfrak{m}} = A$. Also the local ring $C_{\mathfrak{n}}$ dominates $A_{\mathfrak{m}}$. Hence by Exercise 27 we have $C_{\mathfrak{n}} = A$ and therefore $C \subseteq A$.]

5.35

From Exercises 1 and 3 it follows that, if $f : A \to B$ is integral and C is any A-algebra, then the mapping $(f \otimes 1)^* : \operatorname{Spec}(B \otimes_A C) \to \operatorname{Spec}(C)$ is a closed map.
Conversely, suppose that $f : A \to B$ has this property and that B is an integral domain. Then f is integral. [Replacing A by its image in B, reduce to the case where $A \subseteq B$ and f is the injection. Let K be the field of fractions of B and let A' be a valuation ring of K containing A. By (5.22) it is enough to show that A' contains B. By hypothesis $\operatorname{Spec}(B \otimes_A A') \to \operatorname{Spec}(A')$ is a closed map. Apply the result of Exercise 34 to the homomorphism $B \otimes_A A' \to K$ defined by $b \otimes a' \mapsto ba'$. It follows that $ba' \in A'$ for all $b \in B$ and all $a' \in A'$; taking $a' = 1$, we have what we want.]

Show that the result just proved remains valid if B is a ring with only finitely many minimal prime ideals (e.g., if B is Noetherian).[Let $\mathfrak{p}_i$ be the minimal prime ideals. Then each composite homomorphism $A \to B \to B/\mathfrak{p}_i$ is integral, hence $A \to \prod(B/\mathfrak{p}_i)$ is integral, hence $A \to B/\mathfrak{N}$ is integral (where $\mathfrak{N}$ is the nilradical of B), hence finally $A \to B$ is integral.]